

Development, survival, and structural alterations of embryos and alevins of Atlantic salmon, *Salmo salar* L., continuously exposed to alkaline levels of pH, from fertilization

P. G. DAYE¹ AND E. T. GARSIDE

Department of Biology, Dalhousie University, Halifax, N.S., Canada B3H 4J1

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Embryos of the Atlantic salmon, *Salmo salar* L., were incubated continuously from fertilization at pH 6.8 (control) and pH 9.0 and 9.5, initially at 6.7°C but with a gradual decline in the first 5 weeks to 5.0°C for the remaining 10 weeks of exposure. Subsequently, the alevins were maintained in these environments for 50 days after hatching. Developmental processes and hatching were not affected by these levels of pH. Percentage cumulative mortality of treated embryos, 8%, was approximately that in the controls. Alevin mortality in the control lots was 1.2 and 1.3%. At pH 9.0, cumulative mortality was 0.4%, but at pH 9.5 there was an accelerating increase to 18% at the termination of observation.

Sublethal changes in embryos were confined mostly to cell necrosis and sloughed rudimentary epidermis. Some metaplasia of the brain stem occurred at pH 9.5. Sites and intensity of alterations increased in alevins at pH 9.5, following the loss of the zona radiata. In addition to ongoing injury of epidermis, including mucous cells, deleterious alterations occurred in branchial epithelium, erythrocytes, myocardium, blood vessels of the viscera, liver, brain, and optic lenses. In general, sublethal changes caused by hydroxyl ions are similar to those caused by excessive hydrogen ions but are somewhat less extensive in the structures affected or in their degree of severity.

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Des embryons de saumons de l'Atlantique, *Salmo salar* L., ont été gardés en culture dès la fécondation à un pH de 6.8 (témoins) et à des pH de 9.0 et 9.5; la température, d'abord de 6.7°C, a été diminuée graduellement pendant les 5 premières semaines jusqu'à 5°C, température maintenue stable durant les 10 dernières semaines de l'expérience. Après l'éclosion, les alevins ont été gardés dans ces mêmes milieux pour une période de 50 jours. Le développement des embryons et l'éclosion ne sont pas affectés par les pH expérimentaux. Le pourcentage cumulatif des mortalités est à peu près le même (8%) chez les embryons traités et chez les témoins. La mortalité des alevins-témoins a été de 1.2 et 1.3%. A un pH de 9.0, la mortalité cumulative était de 0.4%, mais à un pH de 9.5, la mortalité s'est mise à augmenter de façon accélérée jusqu'à 18% à la fin de l'expérience.

Le pH élevé entraîne quelques changements sublétaux, notamment des nécroses cellulaires et le rejet de l'épiderme rudimentaire. Il peut également se produire une métaplasie du bulbe rachidien à un pH de 9.5. Au pH de 9.5, l'étendue et l'intensité des dommages augmentent chez les alevins après la perte de la zona radiata. En plus des blessures à l'épiderme, y compris aux cellules muqueuses, le pH élevé cause des dommages à l'épithélium branchial, aux érythrocytes, au myocarde, aux vaisseaux sanguins des viscères, au foie, au cerveau et au cristallin. En général, les changements sublétaux causés par les ions hydroxyl sont semblables à ceux causés par un excédent d'ions hydrogène mais ils sont ordinairement moins étendus et moins graves.

[Traduit par le journal]

Introduction

Changes in environmental levels of pH that would affect adversely the development and survival of salmonids generally and the Atlantic salmon, *Salmo salar* L., particularly, in the northern hemisphere, would tend to be those of increasing acidity. The locations of the major breeding ranges of

Atlantic salmon in northeastern North America and northwestern Europe relative to the principal industrial and mining regions and the prevailing movements of the major hemispheric air masses, together with the geologic characteristics of those ranges, have caused and maintained an overall increase in the acidity of much of this range (Beamish *et al.* 1975; Daye and Garside 1979; Jensen and Snekvik 1972). Although the potential for a drastic shift in the pH of such waters is rather low, strongly alkaline bases are common in the effluents of man-

¹Present address: Fisheries and Oceans Canada, Fisheries and Marine Service, Biological Station, St. Andrews, N.B., Canada E0G 2X0.

ufacturing industries and large quantities do reach at least some of the freshwater range of salmon by either deliberate or inadvertent release (Cairns *et al.* 1972; McKee and Wolf 1963).

The effects of alkaline levels of pH on fish have not been investigated extensively. Tolerance levels ranging from pH 9.2 to 10.8 have been stated for a few species of nonsalmonids (Bandt 1936; Carter 1964; Jordan and Lloyd 1964; Trama 1954), and tolerances ranging from pH 9.5 to 9.8 have been given for three salmonids (Carter 1964; Daye and Garside 1975; Jordan and Lloyd 1964; Witschi and Ziebell 1979). However, the only extensive report of sublethal structural effects of basic levels of pH on fish is that given by Daye and Garside (1976). They determined that there was widely distributed injury to surficial tissues of the gill and eye and the integument of underyearling brook trout in exposures of 10 000 min, at pH 9.5. However, no reports have come to notice in which the effects of basic levels of pH have been described for embryonal development and yolk-dependent alevins, in any species.

The present investigation contains a survey of developmental progress, mortality, and structural alterations caused by continuous exposure of embryos and subsequent alevins of Atlantic salmon to alkaline levels of pH 9.0 and 9.5. These values were chosen on the basis of information in a review by the European Inland Fisheries Advisory Committee (EIFAC 1969), which indicated that levels of pH above 9.5 would be lethal to salmonids and that levels below 9.0 would probably not have any detectable effects.

Materials and methods

This study was conducted together with that for the development of Atlantic salmon embryos and alevins at acidic levels of pH, recently reported by Daye and Garside (1979, 1980). The 2400 freshly fertilized salmon ova used in this study were obtained and transported from the Department of Fisheries and Environment fish culture station at Coldbrook, Kings County, Nova Scotia, in the conditions described for the ova used in the other study. These ova were divided into four equal lots at the aquarium, Department of Biology, Dalhousie University. Two lots were incubated continuously in flowing dechlorinated aerated fresh water at an average pH level of 6.8 (range: 6.6–6.9). One of the remaining lots was incubated in water from the same source with the level of pH adjusted to 9.0, and the other lot was incubated similarly, but with the level of pH adjusted to pH 9.5.

The apparatus used for incubation of these ova has been described in considerable detail by Daye and Garside (1979). The alkaline levels of pH were maintained by continuous administrations of a 0.05 *N* solution of sodium hydroxide through precision metering valves in amounts empirically determined to yield pH 9.0 and 9.5 in the incurrent water of the incubation chambers. Daily inspection of the water flowing through the incubation chambers revealed no departures in pH greater than 0.08 units. Temperature of the water in the incubation system

was 6.7°C at the outset and was lowered gradually over 5 weeks to 5.0°C for the remaining 16 weeks.

Progress of embryogenesis was determined by comparison of the lengths of the succession of intervals required to achieve the 18 prehatching structural stages defined by Garside (1959) in the experimental lots with those in the control lot.

Death of an embryo was considered to have occurred when, in early stages, the yolk became opaque white, and, in later stages, the embryo became white. Cessation of opercular movements was the criterion of alevin death. Dead embryos and alevins were removed daily and their numbers were recorded. Samples of live ova and alevins that were removed periodically for the determination of developmental stage or for histologic procedures were also recorded. During the hatching phase, the number that hatched each day was recorded. The percentage cumulative mortality for embryonal and alevin phases were calculated from the computer program BMD 01S (Dixon 1973). Similarly, the percentage cumulative daily hatch of total hatch was calculated for these levels of pH.

Histologic examinations were performed on samples of five embryos at stages 6, 8, 10, 12, 14, 16, and 18 as determined from the series described by Garside (1959) and further elaborated by Daye and Garside (1980), and on samples of five alevins 7 and 28

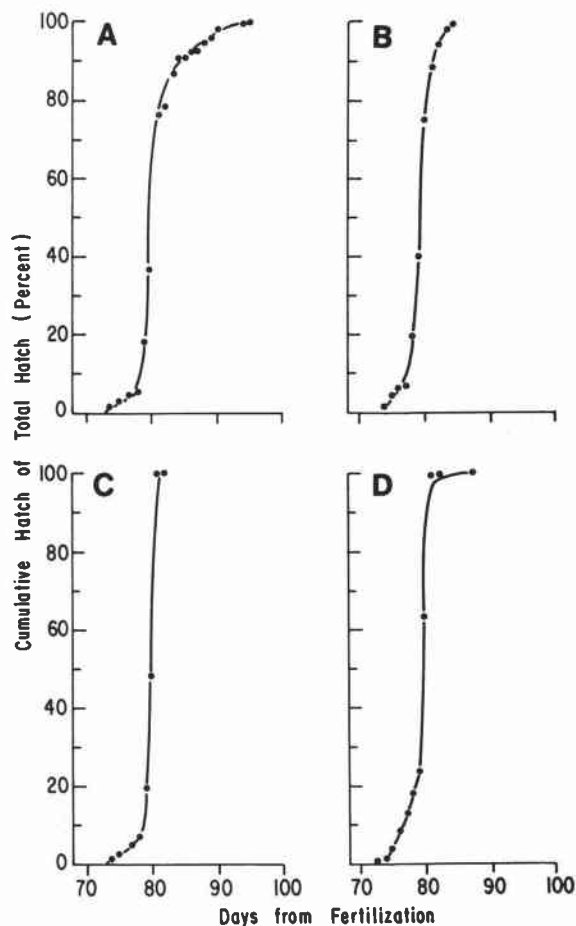


FIG. 1. Daily percentage cumulative hatch of total hatch of embryos of Atlantic salmon reared at a temperature declining from 6.7 to 5.0°C, at levels pH 6.8 (A, B), pH 9.0 (C), pH 9.5 (D).

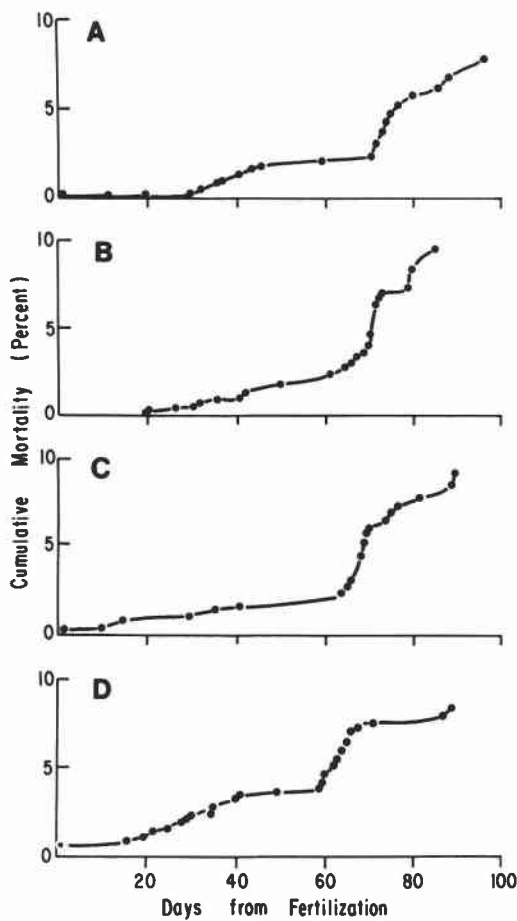


FIG. 2. Daily percentage cumulative mortality of embryos of Atlantic salmon reared at a temperature declining from 6.7 to 5.0°C from fertilization at levels pH 6.8 (A, B), pH 9.0 (C), pH 9.5 (D).

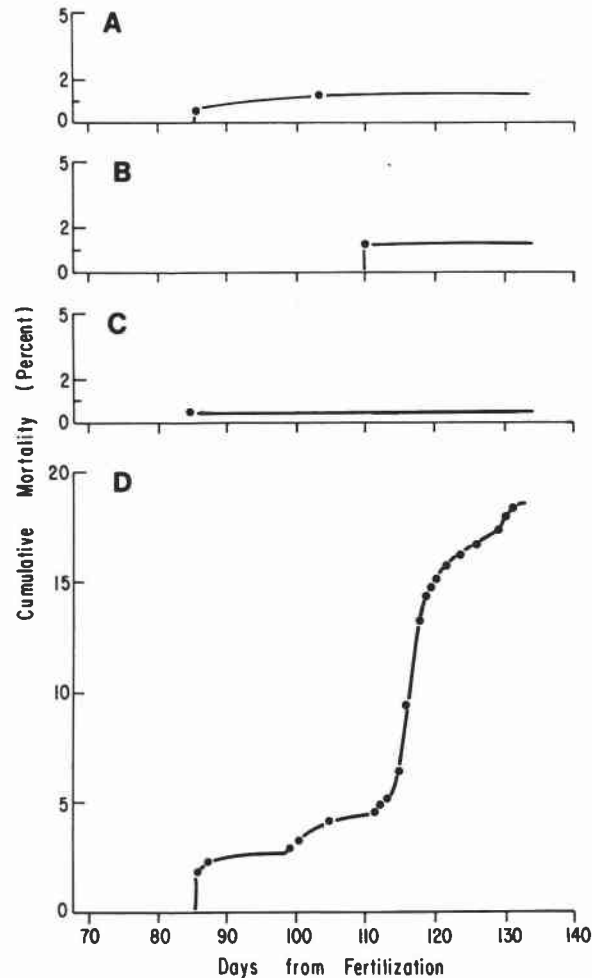


FIG. 3. Daily percentage cumulative mortality of alevins of Atlantic salmon reared at a temperature declining from 6.7 to 5.0°C at levels pH 6.8 (A, B), pH 9.0 (C), pH 9.5 (D).

days after the median hatch, from each of the experimental lots and the control lot. Such examinations were also performed on samples of dead embryos and alevins of variable sample size, randomly drawn from the daily collection, at each level of pH.

Procedures for the preparation of sampled embryos and alevins for histological examination have been described in detail by Daye and Garside (1980). The photomicrographs were prepared only of those structural alterations that were typical in other subjects in the sample, at the same developmental stage, in the same experimental conditions.

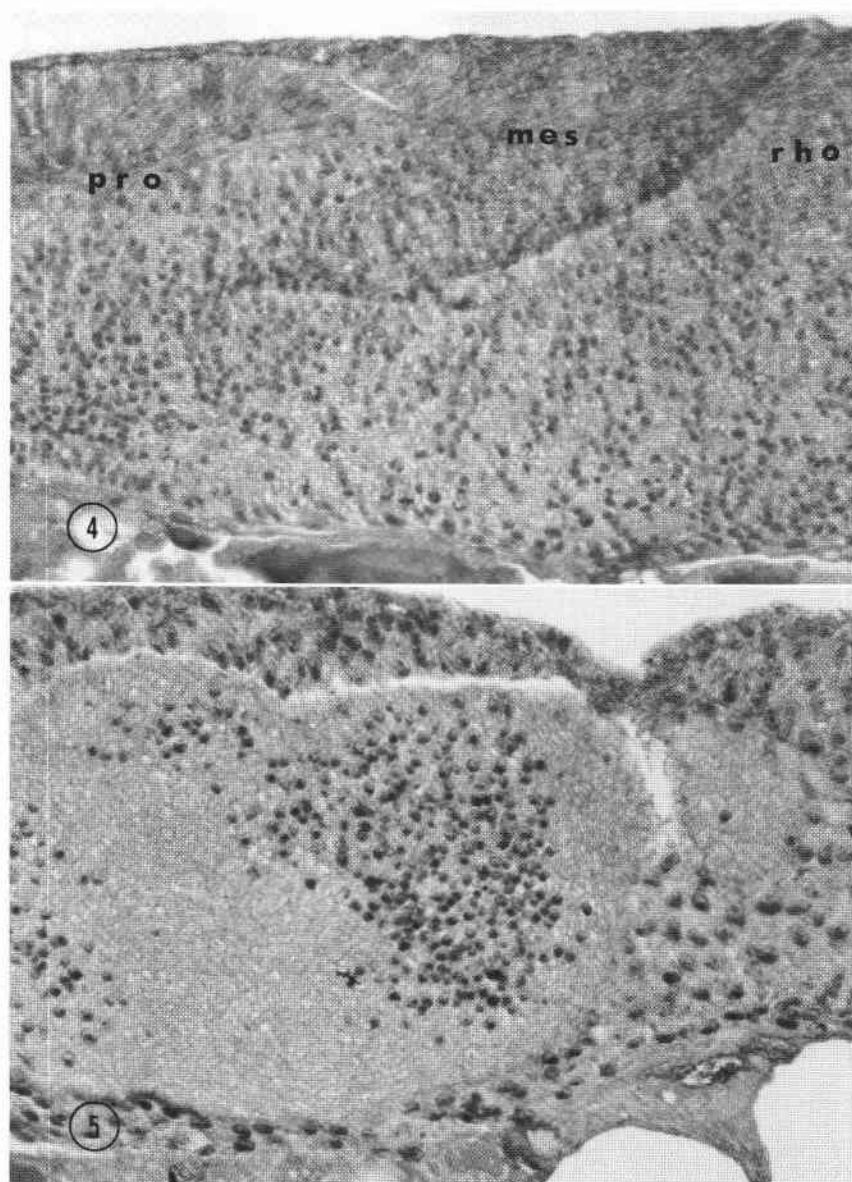
Results

Progress in development, expressed as time in days from fertilization required for embryos in these lots to attain 18 designated prehatching stages, was not affected by the level of pH. The median hatch occurred 80 and 81 days after fertilization in the two control lots incubated at pH 6.8. The median hatch occurred on days 81 and 80 for those lots incubated at pH 9.0 and 9.5, respectively.

Daily percentages of cumulative hatch for each of these lots are presented in Fig. 1.

Embryos incubated from fertilization to the median hatch in the two control lots had cumulative mortalities of 7.9 and 9.6%. Those incubated at pH 9.0 and 9.5 had cumulative mortalities of 8.7 and 8.4% respectively (Fig. 2). Alevins reared for 50 days after their respective day of median hatch had cumulative mortality of 1.2 and 1.3% in the control lots and 0.4 and 18.4% in the lots maintained at pH 9.0 and 9.5 respectively (Fig. 3).

Several microanatomical details in normal development have been described by Daye and Garside (1980) for the seven embryonal stages surveyed in the study concerned with structural alterations caused by acidic levels of pH in salmon embryos. Since normal embryonal structure was the same in the present control lots incubated in the



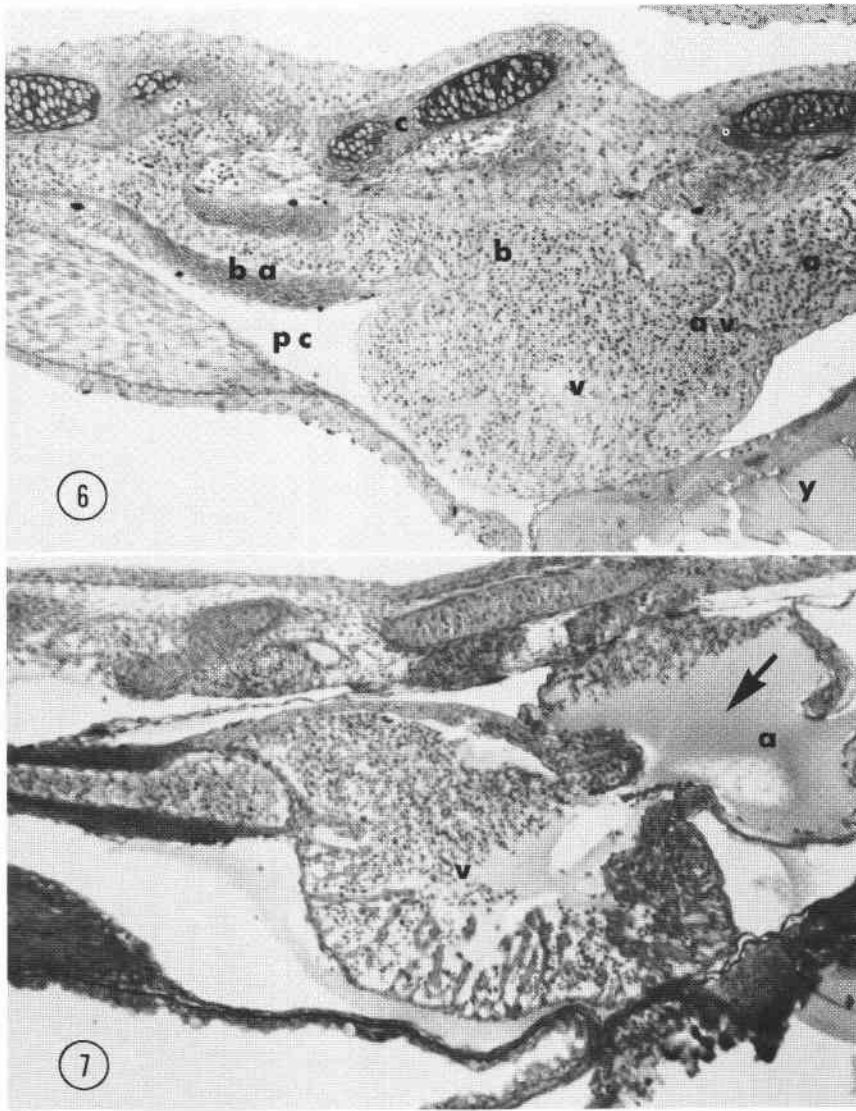
FIGS. 4, 5. Photomicrographs of sagittal sections of the rudimentary brain stem from live embryos of Atlantic salmon at stage 6. Haematoxylin and eosin stain ($\times 720$). Abbreviations: *mes*, mesencephalon; *pro*, prosencephalon; *rho*, rhombencephalon. Fig. 4. Incubated at pH 6.8 (control). Normal. Rudimentary brain stem with discernible mesencephalic, prosencephalic, and rhombencephalic regions. Fig. 5. Incubated at pH 9.5. Atypical metaplasia in all regions of the rudimentary brain stem.

same conditions, such extensive descriptions will not be repeated. Also, in the descriptions that follow, only those apparently pathologic changes of frequent occurrence in the samples from the two experimental lots will be described, illustrated, and compared with the corresponding control status.

Cloudy swelling, cell necrosis, and sloughing occurred over the anterior ectoderm of embryos at stage 6 in both experimental lots but were more

prevalent in the lot at pH 9.5. The three primary divisions of the brain stem had atypical metaplasia in the embryos incubated at pH 9.5 (Figs. 4, 5). Segmentation of lateral mesoderm into somites was more advanced in the lot at pH 9.5 than that in the control lots.

In stage 8, the development of embryos at pH 9.0 was similar to that of the control lots. However, those incubated at pH 9.5 had some hypoplasia of



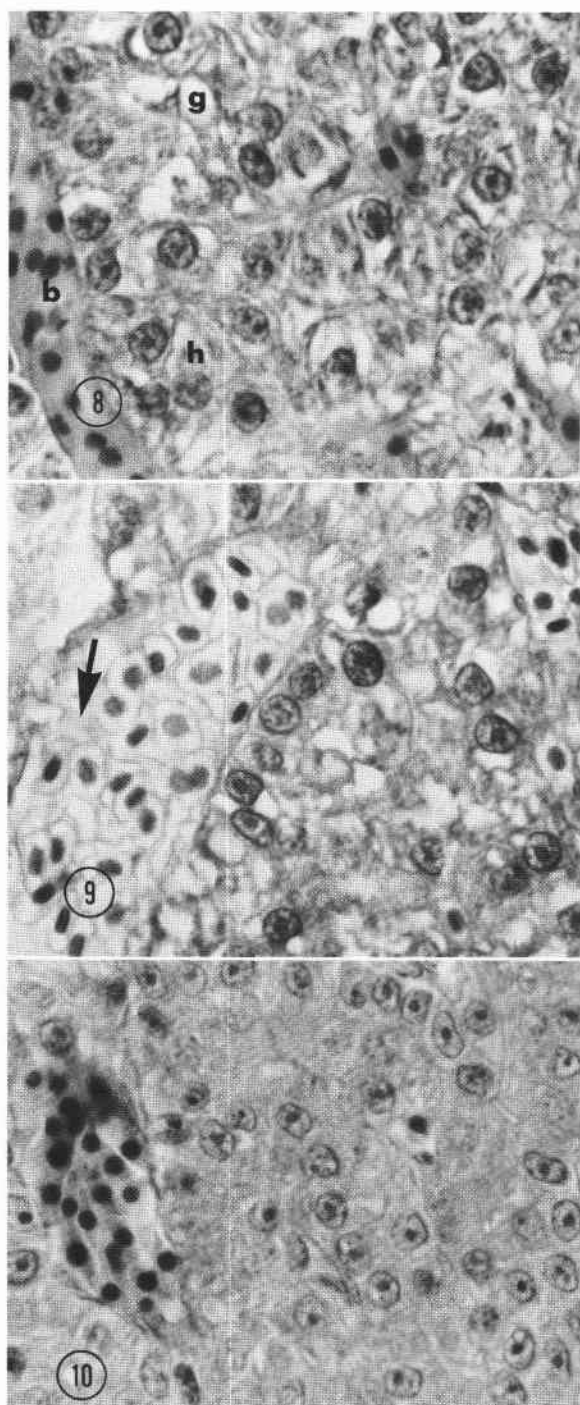
FIGS. 6, 7. Photomicrographs of sagittal sections of the heart from live alevins of Atlantic salmon. Haematoxylin and eosin stain ($\times 288$). Abbreviations: *a*, atrium; *av*, atrioventricular valve; *b*, erythrocytes; *ba*, bulbous arteriosus; *c*, cartilage; *pc*, pericardial cavity; *v*, ventricle; *y*, yolk. Fig. 6. Incubated at pH 6.8 (control). Normal heart, filled with erythrocytes. Fig. 7. Thin myocardium; pericardial effusion; heart filled with transudate (arrow).

the terminal node of the trunk in addition to the continuing necrosis and sloughing of ectodermal cells.

Embryos incubated at pH 9.0 and 9.5 had rather less enlargement of the three divisions of the brain than that of the normal developmental state in the controls, in stage 10. Blood cells were judged by visual appraisal to be more numerous in the control embryos than they were in those incubated at pH 9.0 and 9.5. Dysplastic erythrocytes occurred in the pronephric rudiment in the lots incubated at pH 9.0 and 9.5.

These alterations in normal development in the two experimental lots continued through the remainder of the embryonal term. The only additional change from the normal state was the formation of a transudate within the heart, in stage 16, in the embryos incubated at pH 9.0 and 9.5. This transudate persisted through the alevin stage in both lots.

Several additional alterations occurred in the alevin stage. The extent of such alterations or injuries was invariably greater in the embryos that were incubated at pH 9.5. Only those alevins in the



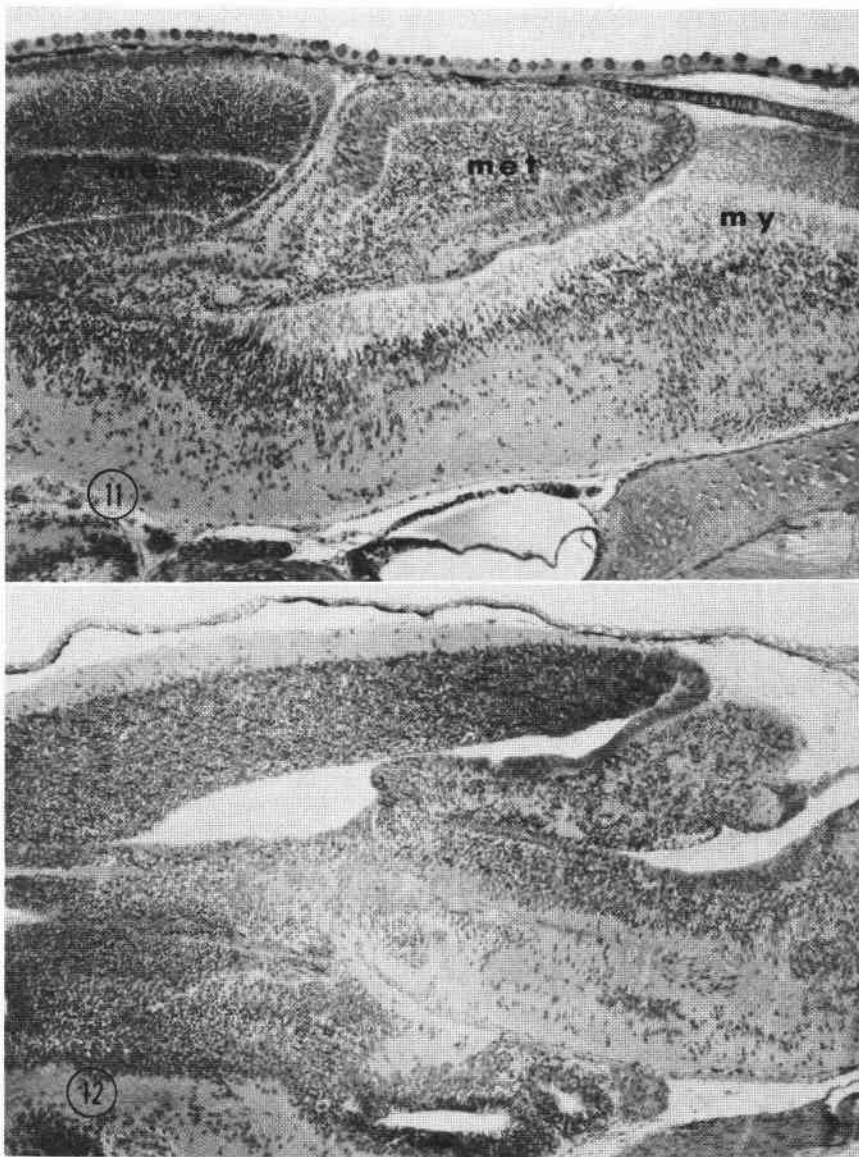
FIGS. 8–10. Photomicrographs of transverse sections of the liver from alevins of Atlantic salmon. Haematoxylin and eosin stain ($\times 1800$). Abbreviations: *b*, erythrocytes; *g*, glycogen; *h*, hepatic cell. Fig. 8. Incubated at pH 6.8 (control). Normal. Hepatic cells with abundant vesicles of glycogen. Fig. 9. Incubated at pH 9.5. Hepatic cells with abundant vesicles of glycogen. Dysplastic erythrocytes. Transudate in blood vessel (arrow). Fig. 10. Incubated at pH 9.0. Dead. Glycogen in hepatic cells almost depleted. Erythrocytes normal.

control lots had normal integument. Some surficial epidermis had been sloughed and mucous cells were void of mucus in alevins reared at both pH 9.5 and 9.0. Although there was no injury to the naris in samples from live alevins, there was some sloughing of olfactory epithelium in alevins that died at pH 9.0, and additionally, there was extensive vesiculation, cell necrosis, and nuclear pyknosis of this epithelium in those that died at pH 9.5. There was dilation of blood vessels in the branchial filaments and lamellae at both alkaline levels of pH. In contrast, alevins examined after treatment and death had constricted blood vessels and epithelial dysplasia. Capillaries in the pseudobranch of surviving and dead alevins reacted in correspondingly similar ways. The myocardium of alevins reared as controls at pH 6.8 was well developed and all chambers were filled with normal blood cells (Fig. 6). However, alevins reared at pH 9.0 and 9.5 had pericardial effusion, excessive transudate, apparently fewer erythrocytes, and thin atrial and ventricular myocardial walls (Fig. 7). Dysplastic erythrocytes and excessive transudate were observed in the blood vessels and sinusoids of the liver, spleen, and kidney, in the alevins reared at pH 9.5 (Figs. 8, 9). Erythrocytes were apparently normal in both surviving and dead alevins reared at pH 9.0 (Fig. 10). There were apparent decreases in liver glycogen in alevins that died during exposure to pH 9.0 and 9.5. Brain structure and neurocyte integrity were normal in alevins incubated at pH 6.8 and 9.0 (Figs. 11, 13). However, the brain of alevins reared at pH 9.5 was metaplastic, particularly so in the mesencephalon and metencephalon (Fig. 12). Neurocytes in the myelencephalon had hypertrophic nuclei (Fig. 14). Dead alevins from all levels of pH, including the controls, had necrosis and nuclear pyknosis occurring commonly through all regions of the brain (Fig. 15). Normal optic lenses were present in all samples of live and dead alevins reared at pH 6.8 and 9.0. However, the lens epithelium was sloughed and the fibers were anaplastic in the alevins reared at pH 9.5. No other parts of the eye were affected by exposure to these levels of alkalinity.

No structural alterations were observed in any other organ or its constituent cells. The otoliths, cartilaginous endoskeleton, notochord, tongue, pharyngeal epithelium, esophagus, gall bladder, intestine, and gonads appeared to be intact and structurally normal in alevins after 30 days of continuous exposure to pH 9.0 and 9.5 following exposure during the entire embryonal phase to these levels.

Discussion

The lack of any detectable controlling effect by

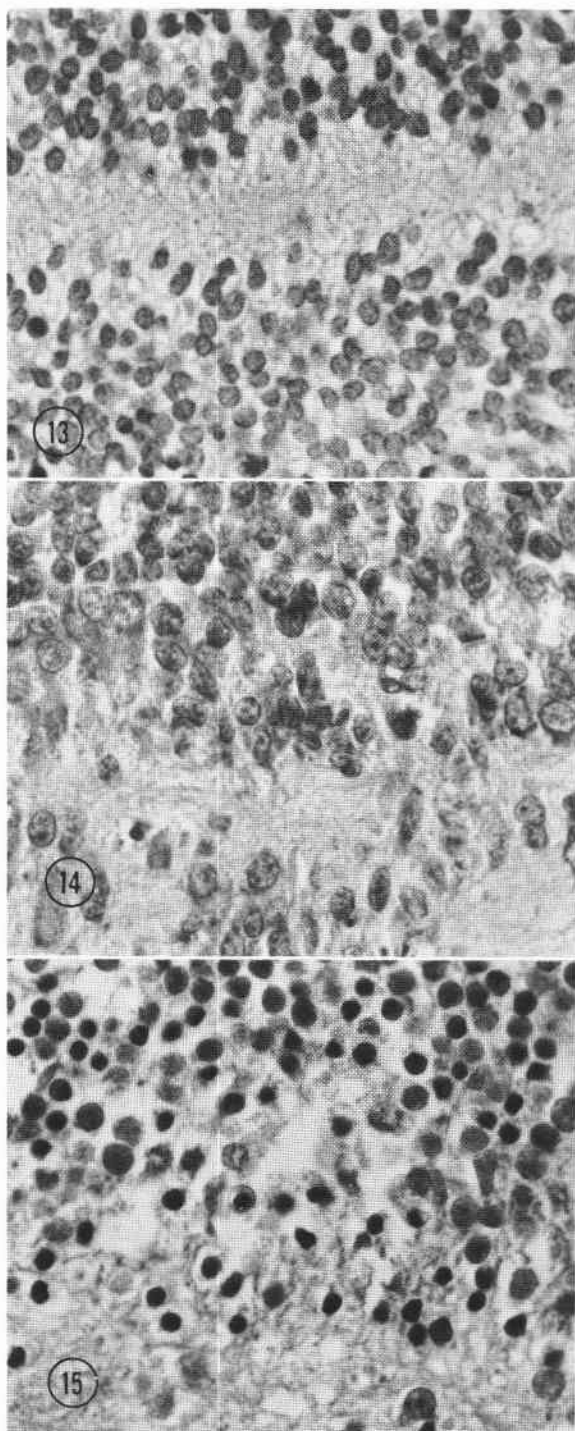


FIGS. 11, 12. Photomicrographs of sagittal sections of the brain from live alevins of Atlantic salmon. Haematoxylin and eosin stain ($\times 225$). Abbreviations: *mes*, mesencephalon; *met*, metencephalon; *my*, myelencephalon. Fig. 11. Incubated at pH 6.8 (control). Normal. Regions of the brain well differentiated. Fig. 12. Incubated at pH 9.5 Extensive metaplasia particularly of mesencephalon and metencephalon.

excessive hydroxyl ions on the general pace of cell division and the degree of cell differentiation during embryogenesis parallels the lack of effect that was observed when salmon embryos were exposed continuously to excessive concentrations of hydrogen ions (Daye and Garside 1979). Such lack of effect suggests that there was no broadly distributed stress that would be manifested in an overall retardation of physiologic processes in the embryo such as that engendered by lowered temperature (Garside 1959, 1966) or reduced concentration of

dissolved oxygen (Garside 1959, 1966; Shumway *et al.* 1964; Silver *et al.* 1963).

The estimate by the European Inland Fisheries Advisory Committee (1969), based on minimal environmental and empirically derived evidence, that basic pH levels from 9.0 to 9.5 would affect the Atlantic salmon adversely has been confirmed in this study. Although neither of the experimental levels of pH can be considered to have been a lethal concentration for embryos, the accelerating mortality of alevins in the last part of the 50-day post-



hatching period indicates that the level of pH that would kill at least half of them was only very slightly higher. Daye and Garside (1975) estimated that the upper mean lethal limit for fingerling brook trout, *Salvelinus fontinalis*, exposed for 10 000 min was pH 9.8.

The zona radiata provided an element of protection from the destructive effects of hydroxyl ions, for these embryos. Although tissue alterations and injuries were greater both in diversity and intensity in naked alevins, relative to those in the embryos, there was, however, no obvious relation between presence or absence of the zona radiata and the extent of mortality. For embryos, the survival value of the zona radiata in relation to abnormally high levels of hydrogen ions is somewhat more apparent (Daye and Garside 1977, 1979, 1980).

In general, the pathologic changes in both embryos and alevins, caused by hydroxyl ions, are similar to those that occurred in these life-stages in response to the exposures to weaker concentrations of hydrogen ions (Daye and Garside 1980). Probably then, higher concentrations of hydroxyl ions would have caused additional parallel changes.

Acknowledgements

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FIGS. 13-15. Photomicrographs of sagittal section of the mid-region of the myelencephalon from alevins of Atlantic salmon. Haematoxylin and eosin stain ($\times 1440$). Fig. 13. Incubated at pH 6.8 (control). Normal. Fig. 14. Incubated at pH 9.5. Hypertrophic nuclei. Fig. 15. Incubated at pH 9.0. Dead. Cellular necrosis and nuclear pyknosis.

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